Project Design Phase-II
Data Flow Diagram & User Stories

Date	15-10-2026
Team ID	23891A05C2
Project Name	Medical Diagnosis Center (ServiceNow Service Portal)
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Level 0 – Context Diagram

```
[Patient] ── registration / test booking / report request ──►
                (Medical Diagnosis Center System)
[Patient] ◄── booking confirmation / appointment status / reports ──

[Receptionist / Lab Technician / Administrator] ── patient, test, appointment,
report and approval updates ──►  (Medical Diagnosis Center System)
[Receptionist / Lab Technician / Administrator] ◄── work queues, records,
status details and dashboard metrics ──
```

DFD Level 1 – Main Processes and Data Stores

```
1. Patient Registration ── patient details ──►  [D1 Patient Records]
2. Test Catalog & Booking ── selected test / date / slot ──►
   [D2 Diagnostic Tests] + [D3 Appointment Records]
3. Booking Workflow ── approval and status updates ──►  [D3 Appointment Records]
   └── confirmation / status notification ──►  Patient
4. Report Management ── results / report file ──►  [D4 Report Records]
   └── report availability / download ──►  Patient
5. Administration & Analytics ── test, booking and report data ──►
   Dashboard for administrators
```

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Optional phase: Payment Processing ── payment details / receipt / refund ──►
[D5 Payment Records]
```

Example: DFD Level 0 (Industry Standard)

User Stories

Use the below template to list all the user stories for the product. Project scope is based on the Medical Diagnosis Center requirements document. Payment is marked optional/phase-based.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Patient	Registration	USN-1	As a patient, I can register and maintain my profile with name, contact details, email and date of birth.	Required patient details are validated and saved; the patient record can be retrieved.	High	Sprint-1
Patient	Diagnostic Test Catalog	USN-2	As a patient, I can view and search active diagnostic tests by name or category.	Only active tests are displayed with category, description, price and duration.	High	Sprint-1
Patient	Appointment Booking	USN-3	As a patient, I can select a diagnostic test, date and available time slot to book an appointment.	Mandatory fields are validated and a booking record with a unique appointment number is created.	High	Sprint-1
Patient	Booking Validation	USN-4	As a patient, I want unavailable or overlapping slots to be blocked so I do not make a duplicate booking.	The system checks slot availability and prevents conflicting bookings before submission.	High	Sprint-1
Patient	Appointment Management	USN-5	As a patient, I can view my appointments and their current statuses in the portal.	The patient can see their own appointment records and statuses such as	High	Sprint-1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
				requested, confirmed, completed or cancelled.		
Patient	Cancellation	USN-6	As a patient, I can cancel an eligible appointment.	The appointment status is updated; where cancellation charges apply, the refund amount is calculated and recorded.	Medium	Sprint-2
Patient	Notifications	USN-7	As a patient, I receive an email confirmation and status updates for my appointment.	Configured notifications are triggered for booking confirmation and relevant status changes.	High	Sprint-2
Lab Technician	Report Management	USN-8	As a lab technician, I can update test processing status and add the diagnostic report for an appointment.	Authorized staff can update report details and attach a report file to the correct patient and appointment.	High	Sprint-2
Patient	Report Access	USN-9	As a patient, I can view and download reports made available for my appointments.	Only reports associated with the patient are shown, and the report is accessible when available.	High	Sprint-2
Receptionist	Appointment Operations	USN-10	As a receptionist, I can review appointment records and help manage booking requests.	The receptionist can find appointment details and update permitted fields according to assigned access.	Medium	Sprint-2
Administrator	Approval Workflow	USN-11	As an administrator, I can approve or reject appointments that require approval.	Approval status, approver and rejection reason are recorded, and the appointment status is updated.	High	Sprint-2
Administrator	Dashboard & Reports	USN-12	As an administrator, I can monitor booking volume, test demand and report status.	Dashboard summaries display available operational counts and trends from system	Medium	Sprint-3

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
				records.		
Administrator	Access Control	USN-13	As an administrator, I can manage role-based access for patients, receptionists and lab staff.	Users can access only the pages, records and actions allowed by their assigned roles.	High	Sprint-3
Patient	Payments (Optional Phase)	USN-14	As a patient, I can view test charges and payment status when payment functionality is enabled.	The system records payment status and can provide a receipt; this feature is delivered only if included in scope.	Low	Sprint-3